1. Find the rate of change of f with respect to t along the curve.

   (a) $f(x,y) = x^2y$, $\mathbf{r}(t) = e^t\mathbf{i} + e^{-t}\mathbf{j}$.

   (b) $f(x,y) = x - y$, $\mathbf{r}(t) = at\mathbf{i} + b\cos at\mathbf{j}$.

   (c) $f(x,y) = \arctan(y^2 - x^2)$, $\mathbf{r}(t) = \sin t\mathbf{i} + \cos t\mathbf{j}$.

2. Find the stationary points and the local extreme values.

   (a) $f(x,y) = (x^2 + y^2)e^{x^2-y^2}$.

   (b) $f(x,y) = x^2 + 2xy + 3y^2 + 2x + 10y + 1$.

   (c) $f(x,y) = \frac{x}{y} - \frac{y}{x}$.

   (d) $f(x,y) = \cos x \cosh y$, $-2\pi < x < 2\pi$.

3. Let $f(x,y) = x^2 + kxy + y^2$, k a constant.

   (a) Show that f has a stationary point at $(0, 0)$ no matter what value is assigned to k.

   (b) For what values of k will f have a saddle point at $(0, 0)$?

   (c) For what values of k will f have a local minimum at $(0, 0)$?

   (d) For what values of k is the second-partials test inconclusive?

4. Find the absolute extreme values taken on by f on the set indicated.

   (a) $f(x,y) = (x-3)^2 + y^2$; $0 \le x \le 4$, $x^2 \le y \le 4x$.

   (b) $f(x,y) = x^2 - 2xy + y^2$; $0 <\le x \le 6$, $0 \le y \le 12 - 2x$.

   (c) $f(x,y) = x^2 + 4y^2 + x - 2y$; the closed region enclosed by the ellipse $\frac{1}{4}x^2 + y = 1$.

5. Minimize $x^2 + y^2$ on the hyperbola $xy = 1$.

6. Maximize xy^2 on the ellipse $b^2x^2 + a^2y^2 = a^2b^2$.